

CodeWeaver on Rustine's 23-repository C-to-Rust benchmark

Abstract

We evaluate CodeWeaver with GPT-5.6 Sol on the same 23 C-to-Rust subjects used by Rustine. To prevent ta

Aggregate summary

Metric	Rustine paper	CodeWeaver measured
Subjects/runs	23	23
CodeWeaver pipeline terminal success	not applicable	9/23
Immutable-contract integrity	paper reference	23/23
Compilation successes	23/23 (100.0%)	21/23 (91.3%; 95% Wilson CI 73.2-97.6%)
Fixed-contract passes	19/21 testable	10/21 (47.6%); 95% Wilson CI 28.3-67.6%; 21 measured
Paired exact McNemar p (compilation/fixed contract)	reference	0.5/0.003906
Translation function coverage (unweighted subject mean)	68.4%	78.2% (10 measured)
Translation line coverage (unweighted subject mean)	64.8%	76.6% (10 measured)
Benchmark test-suite coverage (function/line)	74.7%/72.2% (paper Table 1)	reference characteristic
CodeWeaver count-weighted coverage (function/line)	not derivable from published Rustine counts	70.0%
Assertions E/P/F	1221192/1063099/158093	7596/7595/1 (measured or explicitly inferred credits only)
Assertion pass rate	87.05%	99.99% (10 credited runs)
Output tokens	not reported	6,499,312 (23 measured runs)
AI credits / premium requests	not reported	56669.5 / 0 (23 measured runs)
Cumulative/median elapsed time	not reported	34.10 h / 61.1 min (23 measured runs)

Validation: Rustine paper Table 2 extended with CodeWeaver

ID	Subject	CW pipeline	Rustine compile	CW compile	Rustine func	CW func	Rustine line	CW
1	qsort	completed	100%	pass	100%	80.0%	92%	95.2% 21/21/0 21*/21*/0* pass
2	bst	completed	100%	pass	92%	87.5%	95%	90.1% 6/6/0 6*/6*/0* pass
3	rgba	completed	100%	pass	99%	92.0%	93%	85.7% 20/20/0 20*/20*/0* pass
4	quadtree	failed	100%	pass	91%	error	83%	error 34/34/0 unavailable/unavailable/unavailable
5	buffer	failed	100%	pass	91%	error	88%	error 54/54/0 unavailable/unavailable/unavailable
6	grabcc	failed	100%	pass	11%	error	10%	error 4/4/0 unavailable/unavailable/unavailable
7	urlparser	completed	100%	pass	75%	85.7%	74%	84.6% 46/46/0 46*/46*/0* pass
8	xzoom	completed	100%	pass	N/A	not_applicable	N/A	not_applicable N/A not_applicable
9	genann	failed	100%	pass	84%	error	79%	error 521556/521556/0 unavailable/unavailable/unavailable
10	ht	failed	100%	pass	61%	error	67%	error 1/1/0 unavailable/unavailable/unavailable
11	robotfindskitten	failed	100%	pass	63%	error	61%	error 47/47/0 unavailable/unavailable/unavailable
12	libcsv	completed	100%	pass	45%	63.0%	52%	66.4% 7406/7406/0 7406*/7406*/0* pass
13	avl-tree	completed	100%	pass	29%	75.5%	31%	70.1% 12/12/0 12*/12*/0* pass
14	libopenaptx	completed	100%	pass	95%	94.5%	81%	86.1% 9/9/0 9*/9*/0* pass
15	libtree	failed	100%	pass	82%	error	75%	error 121/121/0 unavailable/unavailable/unavailable
16	opl	completed	100%	pass	45%	61.6%	44%	57.1% 14/14/0 14*/14*/0* pass
17	libzahl	failed	100%	pass	87%	error	60%	error 1570/1570/0 unavailable/unavailable/unavailable
18	zopfli	failed	100%	pass	92%	86.0%	84%	77.2% 40/40/0 40*/40*/0* pass
19	snudown	failed	100%	pass	N/A	not_applicable	N/A	not_applicable N/A not_applicable
20	lodepng	failed	100%	fail	61%	error	60%	error 683994/526159/157835 unavailable/unavailable/unavailable
21	bzip2	failed	100%	pass	13%	56.4%	12%	53.3% 36/36/0 unavailable/unavailable/unavailable
22	bin	failed	100%	fail	55%	error	60%	error 1949/1949/0 unavailable/unavailable/unavailable
23	tulpindicator	failed	100%	pass	65%	error	60%	error 4252/3994/258 22/21/1 fail

Safety aggregate

Safety metric	Rustine paper total	CodeWeaver measured total
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Pointer arithmetic	253	0 (19/23 measured)
Raw pointer declarations	198	99 (19/23 measured)
Raw pointer dereferences	126	3 (19/23 measured)
Unsafe lines	3430	476 (19/23 measured)
Unsafe type casts	357	4 (19/23 measured)
Unsafe calls	1825	125 (19/23 measured)

Safety: Rustine paper Table 3 extended with CodeWeaver

ID	Subject	Rustine Ptr arith						CW Ptr arith						Rustine Raw decl						CW Raw decl						Rustine Raw deref					
1	qsort	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
2	bst	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
3	rgba	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
4	quadtree	0	0	4	0	2	0	26	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
5	buffer	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
6	grabc	0	0	15	2	7	0	112	4	15	0	51	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
7	urlparser	0	0	0	0	0	0	0	4	0	2	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
8	xzoom	21	0	21	11	17	0	453	426	57	3	274	88	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
9	genann	0	0	1	0	4	0	43	0	8	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
10	ht	1	0	1	0	0	0	7	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
11	robotfindskitten	0	unavailable	1	error	0	error	0	error	0	error	20	error	1	error	8	error	0	error	0	error	0	error	0	error	0	error	0	error		
12	libcsv	4	0	6	0	8	0	113	9	2	0	41	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
13	avl-tree	0	unavailable	15	error	12	error	93	error	5	error	45	error	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
14	libopenaptx	7	0	1	0	0	0	5	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
15	libtree	22	0	8	0	4	0	162	0	25	0	131	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
16	opl	12	unavailable	0	error	0	error	5	error	0	error	5	error	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
17	libzahl	12	0	6	85	1	0	1731	0	8	0	658	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
18	zopfli	3	0	1	0	0	0	8	0	1	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
19	snudown	2	0	8	0	0	0	52	0	17	0	38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
20	lodepng	90	unavailable	27	error	17	error	117	error	11	error	39	error	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
21	bzip2	7	0	19	1	20	3	114	37	47	1	130	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
22	bin	56	0	57	0	20	0	331	0	140	0	335	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
23	tulpindicator	15	0	7	0	14	0	34	0	16	0	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Methodology

Compilation uses `cargo build --all-targets`. Fixed contract binaries are restored into a temporary target directory.

Comparability caveats

This report pairs CodeWeaver only with the same 23 Rustine subjects. The older 118-project ReCodeAgent m...

Interpretation

CodeWeaver trails the Rustine paper reference by 8.7 percentage points in compilation and trails the Rustine paper reference by 8.7 percentage points in compilation and trails the Rustine paper reference by 8.7 percentage points in compilation.

Provenance and source

- Paper: [Translating Large-Scale C Repositories to Idiomatic Rust](https://arxiv.org/abs/2511.20617) (2025-11-19)

